Overcoming the Cold-Start Calibration Barrier in Distributed Automotive Radar Re-Simulation via Dual-Horizon Dynamic Alignment and Preflight Seeding

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***Abstract*—Distributed replay must initialize calibration state as well as kinematic tracks. We analyze a dual-horizon alignment schedule, physically qualified observations and provenance-aware preflight seeding, with explicit separation between a proposed estimator and the archived wrapper integration. Exact scalar error products show why a reciprocal gain is an e-folding scale rather than a threshold-convergence guarantee. We derive timestamp deskew, clock-bias identifiability and track-conditioned beat estimation, and document source-level limits. Paired replay counts are reproduced while unverified accuracy and cluster-speedup claims remain clearly identified as reports.**

I Introduction: why replay state is not just tracker state

A distributed replay job has two distinct continuity problems. The object tracker needs sufficient recent measurements to initialize its kinematic state. The radar front end may also carry mounting alignment, clutter statistics, temperature history and qualifier counters. A single warm-up policy is valid only if it covers the slowest state on the path to the scored output. Splitting a 300-log drive across twenty workers gives roughly fifteen scored logs per worker; a preceding log can be replayed without scoring, but its adequacy must be measured in accepted calibration updates rather than file count. The project narrative reports a 16-log chunking attempt and a calibration bottleneck. Its conversion between 400 frames, 30–40 logs and twenty minutes is inconsistent without an independent log-duration definition. At 20 Hz, 400 frames are twenty seconds and 46 frames are 2.3 seconds. We therefore treat the 360,000 dropped-frame and cluster-speedup statements as project reports requiring scheduler and KPI logs, not as measured outcomes established here. The useful research question survives this correction: how should a replay distinguish estimator warm-up from validated state initialization?

I-A System architecture and scope

The Gen8 iND13400 sensor repository contains the BBE32-side signal-processing wrappers and pure C kernels. Its relevant chain is ADC, range FFT, Doppler FFT, CFAR, CDC packing, beamforming, alignment and radar output. The Domain Controller repository contains DC_SIL wrappers and the unified F360 tracker. Preflight header handling and temporal deskew connect these domains. The separate 4D RadarSplat/V&V repository is deferred; no validation result from it is claimed.

II Related work and contribution boundary

Recursive estimation follows Kalman's state-space framework [1]; practical detection-to-track association and motion-model management are established subjects. The proposed schedule is an engineering gain policy, not a new proof of Kalman optimality. The distinct project contribution is a traceable combination of startup scheduling, physically qualified updates, replay-state seeding and guarded integration. The residual-frequency estimator is related to the long-established phase-difference frequency-estimation literature [4]. Neither an exponential schedule nor a stored calibration field alone establishes patent novelty.

III Dual-horizon dynamic alignment

Consider a scalar local alignment observation $z_N = \theta^\star + \eta_N$ and an admitted update with gain K_N:

$$\hat{\theta}_{N+1} = \hat{\theta}_N + K_N(z_N - \hat{\theta}_N). \tag{1}$$

Here $\hat{\theta}_N$ is the estimate, $\theta^\star$ the locally constant mounting angle, and η_N measurement noise. This reduced model is an analysis model; it is not a recovered implementation of the buried production DABB recursion.

$$K_N = K_f e^{-\alpha N} + K_s. \tag{2}$$

The fast component $K_f = 0.05$ and slow component $K_s = 0.0025$ are dimensionless. The archived narrative uses $\alpha = 0.08$ per update; the wrapper and host A/B code use a default of 0.1. These are distinct configurations, not interchangeable measurements. An update index must advance only under the documented eligibility policy; advancing it on rejected scans changes behavior.

$$e_N = e_0 \prod_{i=0}^{N-1}(1-K_i), \quad e_N = \hat{\theta}_N - \theta^\star. \tag{3}$$

In the noiseless model and for $0 < K_i < 1$, error contracts monotonically. Because $\log(1-u) \le -u$ for $0 < u < 1$, a convenient upper bound is

$$\frac{|e_N|}{|e_0|} \le \exp\left[-NK_s - K_f \frac{1-e^{-\alpha N}}{1-e^{-\alpha}}\right]. \tag{4}$$

All terms are dimensionless; this is an upper bound, not an equality or a guaranteed 46-frame lock. A fixed gain 0.0025 has e-folding time $-1/\log(0.9975) \simeq 399.5$ updates. A requested threshold depends on the initial error and noise, not simply on the reciprocal gain.

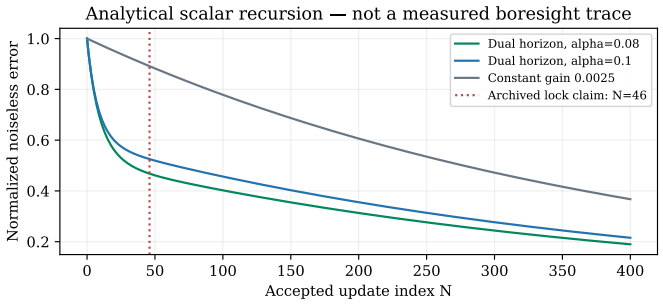


Figure 1: Exact products for the explicitly stated scalar model. The archived frame-46 claim is marked for context, not fitted into the curves.

III-A Noise and lock certificates

For independent zero-mean noise with variance σ_η^2, the scalar covariance obeys

$$P_{N+1} = (1-K_N)^2 P_N + K_N^2 \sigma_\eta^2. \tag{5}$$

The steady fixed-gain variance is $K_s\sigma_\eta^2/(2-K_s)$. A fast initial gain trades noise sensitivity for bias contraction. Ten small successive estimate changes are a stability indicator, not a certificate of absolute alignment: a frozen estimator is stable even if biased. A defensible release test combines observation support, residual statistics, independent reference error where available, and a bounded hold count.

III-B Stationary-clutter qualification

For a stationary world scatterer, a nonrotating sensor moving at ego velocity $\mathbf{v}_e$ predicts radial rate $-\mathbf{u}^T\mathbf{v}_e$. An admissibility gate is

$$|v_r + \mathbf{u}^T\mathbf{v}_e| \le \varepsilon, \quad \mathbf{u} = (\cos\theta, \sin\theta)^T. \tag{6}$$

The radial speed v_r and tolerance ε are in m/s. Sensor lever-arm rotation, sign conventions and aliasing must be handled before this comparison. A reflector can be stationary and therefore legitimately satisfy this gate; the gate does not reject guardrails merely because they are guardrails. No numerical tolerance is invented where the source does not specify one.

IV Preflight seeding and provenance

The header helper accepts azimuth and elevation in degrees within 5 and 2 degrees respectively, with qualifier checks in the transport helper. This validates magnitude and presence, not freshness, sensor identity, finite values or convergence history. Production preflight should bind the seed to sensor serial number, mounting configuration, firmware version, timestamp and quality state. A rejected or stale seed must leave the fallback estimator operational. NaN values require explicit finite checks because ordinary comparisons do not reject them. Historical header injection can avoid estimator initialization delay when the seed is valid. It does not eliminate I/O or replay startup latency, and using a future converged value in a scored online experiment creates look-ahead leakage. Such injection is appropriate for a declared offline replay mode; online cold-start accuracy must be tested without future information.

V Temporal deskew and clock separation

The archived wrapper implements the following explicit coordinate convention:

$$\mathbf{p}_c = R(\omega\Delta t)\mathbf{p} - \mathbf{v}\Delta t - \tfrac{1}{2}\mathbf{a}\Delta t^2. \tag{7}$$

Here position is in metres, velocity in m/s, acceleration in m/s^2, yaw rate in rad/s and signed Δt in seconds. This expression is a specified approximation, not a universal sign convention for every SE(2) time transformation. The rotation maps vectors between the chosen timestamped ego frames; translation must be expressed in a compatible frame. A physical test should use a stationary landmark during known translation and rotation to establish the signs.

$$R(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}. \tag{8}$$

Its first-order timestamp sensitivity is $\omega R'(\omega\Delta t)\mathbf{p} - \mathbf{v} - \mathbf{a}\Delta t$, so long-range points are particularly sensitive to angular-rate uncertainty.

V-A Yaw-free clock model

Under constant target velocity, let the displaced observation vector be $\mathbf{d} = \mathbf{p} - \tau\mathbf{v}$ while contemporaneous range and range rate satisfy $rw = \mathbf{p}^T\mathbf{v}$. Then

$$rw - \mathbf{d}^T\mathbf{v} = \|\mathbf{v}\|^2 \tau. \tag{9}$$

The range r is in m, radial speed w in m/s, and both sides have units m^2/s. Shared yaw rotations cancel in the inner product. A radial-rate bias b gives $y_k = a_k\tau + r_k b$, with $a_k = \|\mathbf{v}_k\|^2$. Two nondegenerate targets give

$$\begin{bmatrix} \tau \\ b \end{bmatrix} = \begin{bmatrix} a_1 & r_1 \\ a_2 & r_2 \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}. \tag{10}$$

Identifiability requires $a_1 r_2 - a_2 r_1 \neq 0$ and adequate conditioning. The archived DC update uses an ego-motion proxy, not paired target observations with these physical units; its outputs are a sidecar demonstration rather than a validated clock estimate.

V-B Relative yaw from common displacements

If two sensors measure the same nonzero world displacement $\Delta\mathbf{p}_k$, their local complex displacements obey $D_{ik} = e^{-j\beta_i}\Delta p_k$. Consequently $\arg\sum_k D_{ik}\overline{D_{jk}} = \beta_j - \beta_i$ in the ideal common-pair case. This estimates relative yaw only. Absolute yaw additionally requires surveyed baseline closure or another external reference; zero baseline, negligible displacement and mismatched pairs are failure cases, not calibration successes.

VI Track-conditioned residual beat estimation

For complex fast-time samples x_n at rate f_s, derotate a predicted beat frequency f_p and form

$$z_n = x_n e^{-j2\pi f_p n/f_s}, \quad C_1 = \sum_{n=1}^{M-1} z_n \overline{z_{n-1}}. \tag{11}$$

$$\hat{f}_{\text{res}} = \frac{f_s}{2\pi}\arg(C_1), \quad \hat{r} = \frac{c(f_p + \hat{f}_{\text{res}})}{2\mu}. \tag{12}$$

Here $f_s, f_p, \hat{f}_{\text{res}}$ are in Hz and FMCW chirp slope μ is in Hz/s. This estimates beat frequency and therefore range under the stated model; it is not automatically a slow-time Doppler estimator. The C helper returns cycles/sample and assumes samples were already derotated: its `f_pred` argument is explicitly unused. The caller owns the phase convention and physical scaling. The separate four-sample cubic stencil computes three adjacent phase products. It is not the same algorithm as arbitrary-length lag-one accumulation. Reported 4.5-fold multiply savings and 0.108 m INT8 RMSE require a specified reference estimator, window length, quantizer, chirp slope and noise distribution; those prerequisites cannot be recovered from a headline alone. Quantization angle bounds require a valid amplitude-to-error ratio, and an arcsine input outside $[-1, 1]$ must be rejected or bounded.

VII Firmware realization and integration status

The exported Gen8 wrapper computes a gain readout and caps update-count configuration at fifty; its own comment says the buried DABB recursion cannot be rewritten by that wrapper. A proposed gain policy is therefore distinguished from a demonstrated production update law. The C kernels are bounded and heap-free at their local interfaces; that observation says nothing about whole-system stack usage or timing. Build definitions connect the new libraries, while compile-time guards select behavioral paths. Guarding code is a rollback mechanism, not evidence of zero deployment risk.

VIII Experimental evidence and its limits

The strongest directly inspectable artifacts are the paired CSV outputs of an instrumented replay harness. They contain scan identifiers, track and detection counts, association counts, assimilation marks and a sum of longitudinal track velocities. They do not contain independently surveyed target trajectories, classification labels, position errors or recall denominators. Accordingly, changes in these counters are observations about the harness, not estimates of fleet accuracy or false-alarm probability. The harness additionally constructs pairing candidates using azimuth and range proximity; such labels are hypotheses rather than truth annotations. The association-count difference is -347, or -3.66%. It is not a verified reduction in false associations, because the CSV files do not distinguish correct and incorrect matches. Marks may be counted at an instrumented update site and should not be equated automatically to distinct reflected targets. Repeated targets and repeated use of a detection must be audited separately. The recorded mean

Table 1: Paired replay ledger, independently summed from CSV files

Quantity	OFF	ON
Replay records	155	155
Valid detections	71,779	71,779
Association events	9,478	9,131
Assimilation marks	0	10,130
Records with marks	0	151

and maximum longitudinal velocity shifts, 0.910 and 3.743 m/s, are changes relative to OFF, not velocity errors relative to truth. The archive reports parent position RMSE 1.969 to 0.651

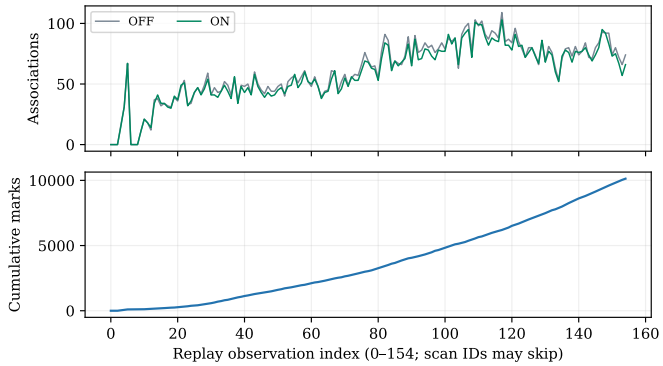


Figure 2: Actual paired replay association series and cumulative ON assimilation marks. No random series or interpolated per-scan observations are used.

m, spray RMSE 0.911 to 0.315 m, fixed/adaptive cut-in recall 86.3/99.6%, and IMM recall 88.0%. Those figures are retained as reported benchmark claims. Their truth trajectories, sample definitions and executable scoring pipelines are not provided by the paired CSVs. They are therefore not demonstrated fleet accuracy results in this manuscript. The reported 67.0% parent-RMSE reduction is the arithmetic $1 - 0.651/1.969$; correct arithmetic does not validate the underlying experiment.

VIII-A Host checks and module-level evidence

The archival ledger records 40/40 host checks for both configurations. Equal pass counts establish that the named checks passed; they do not establish binary equivalence, absence of undefined behavior, timing safety, or functional safety. A strict baseline regression comparison requires matching inputs, reset states, compiler options, output fields and tolerances. The four module probes report identity fill (T1), entropy inflation (T2), assimilation marking (T3), and penalty exclusion (T4). In T2, $0.5\exp(2) = 3.694528$ is a deterministic arithmetic probe, not a wet-road accuracy test.

VIII-B Validation design required for a publication claim

Partition complete drives before parameter tuning. Keep sensor poses, calibration history and overlapping temporal windows within a single split to avoid leakage. Establish

reflector and target ground truth independently of the algorithm under evaluation. Report position and velocity errors in common coordinates, per-target association precision and recall, and duplicate-update rates. Aggregate confidence intervals at drive or target-trajectory level rather than pretending adjacent scans are independent. Publish ablations for gating, reflector covariance, confirmation, seeding and geometry separately. A hardware deployment claim additionally needs the production DSP build and measured worst-case execution time.

IX Conclusion

The project exposes useful estimator and interface design problems, but reproducibility requires a strict separation between equations, implemented call sites and measured evidence. The source supports inspection of guarded kernels and replay counters; it does not by itself support universal lock-time, safety or fleet-accuracy guarantees. The next decisive experiment is an independent ground-truth evaluation with complete configuration and per-drive ablations.

References

[1] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," Journal of Basic Engineering, vol. 82, pp. 35–45, 1960, doi:10.1115/1.3662552.

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[3] H. Rohling, "Radar CFAR thresholding in clutter and multiple target situations," IEEE Transactions on Aerospace and Electronic Systems, vol. AES-19, pp. 608–621, 1983, doi:10.1109/TAES.1983.309350.

[4] S. M. Kay, "A fast and accurate single frequency estimator," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 37, pp. 1987–1990, 1989, doi:10.1109/29.45547.

[5] A-PRISM project, "Master archival backup," one.md, October 2026. Internal source/equation/claim ledger; not an independently reviewed publication.

[6] A-PRISM project, paired instrumented replay CSVs, real_off.csv and real_on.csv. Source files and hashes reproduced in the extended reproducibility appendix.

APPENDIX A: INVENTION DISCLOSURE FORM (1 OF 2)

Status: internal technical disclosure draft; inventors, assignee and filing details are not supplied.

Field of invention

This disclosure concerns sensor alignment and distributed replay initialization in automotive radar software and embedded signal processing.

Background and failure modes

State continuity, coordinate consistency and observation ownership can fail independently. An estimator that appears converged can remain biased; a correct image construction can be consumed in the wrong coordinate frame; a measurement can be counted twice or penalized after assimilation. Existing estimation and image-method techniques are background knowledge, not asserted absent from prior art.

Detailed embodiment

The embodiment combines explicit coordinate and unit conventions, bounded state transitions, provenance-qualified inputs and instrumentation at the production interface. The manuscript defines the mathematical operations; the reproducibility appendix identifies implemented helpers and integration gaps. Proposed checks are not represented as already present in every archived source path.

Commercial detectability analysis

A controlled reset/replay comparison can observe angle-state transients, seed acceptance, measurement ownership and covariance changes. Telemetry alone does not uniquely identify an internal algorithm: alternative designs can produce similar curves. Source inspection, configuration records and reproducible boundary tests are stronger evidence than a single output trace.

Evidence and disclosure boundaries

Reported performance values require their original scoring data. This document does not establish legal novelty, claim validity, infringement or a safety certification. The claim draft deliberately avoids promising a universal lock time or a particular accuracy gain. Inventorship and disclosure dates must be completed from actual project records.

APPENDIX A: CLAIMS SPECIFICATION (2 OF 2)

One independent claim followed by eight dependent claims.

Claim 1 (independent) A method of initializing a radar alignment estimator, comprising selecting eligible stationary observations, determining a scan-dependent gain having a decaying component and a nonzero steady component, updating an alignment state using the eligible observations, and accepting a provenance-qualified historical state to initialize an explicitly identified offline replay mode.

Claim 2 (dependent) The method of claim 1, wherein the decaying component is exponential in an accepted-update index.

Claim 3 (dependent) The method of claim 1, wherein eligibility compares measured radial velocity with an ego-motion projection.

Claim 4 (dependent) The method of claim 1, wherein state provenance identifies a physical sensor and its mounting configuration.

Claim 5 (dependent) The method of claim 1, wherein historical-state acceptance checks finite values, quality, age and angular bounds.

Claim 6 (dependent) The method of claim 1, wherein rejection of the historical state retains an operational cold-start path.

Claim 7 (dependent) The method of claim 1, wherein timestamped satellite observations are transformed into a common coordinate epoch.

Claim 8 (dependent) The method of claim 1, wherein a release qualifier combines residual support with a bounded consecutive-update criterion.

Claim 9 (dependent) The method of claim 1, wherein configuration-controlled comparison records preserve distinct baseline and experimental initialization modes.